MITHUN P

Software Development Engineer | JavaScript • React • Node.js • REST APIs

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PROFESSIONAL SUMMARY

Computer Science Engineering student with a strong foundation in Data Structures & Algorithms, Object-Oriented Programming, and full-stack development. Delivered 2 production-grade projects integrating REST APIs, real-time databases, and automated business logic — independently owned from architecture to deployment. Actively building proficiency in JavaScript (ES6+), React, and Node.js

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), Python, Java, Dart
Frontend:	React.js (Hooks, Component Architecture, State Management), HTML5, CSS3 (Flexbox, Grid)
Backend & APIs:	Node.js, RESTful API Design (CRUD, HTTP methods, status codes), GraphQL (basics), async/await, Promises
Databases:	MySQL, Firebase Realtime DB, SQL (Joins, Indexing, Normalization, Queries)
DSA:	Arrays, Strings, HashMaps/Sets, Linked Lists, Stacks, Queues, Trees, Binary Search, Sorting, Recursion, Big-O Analysis
CS Fundamentals:	OOP, DBMS, Operating Systems, Computer Networks
Tools & Practices:	Git (Branching, PRs, Commit hygiene), GitHub, VS Code, Postman, Android Studio

PROJECT EXPERIENCE

AI-Powered Note Summarization App — Full-Stack Application

JavaScript • REST APIs • Firebase • Component-Based UI | 2025

- Architected and delivered a full-stack application integrating 2+ third-party NLP/AI REST APIs; implemented async/await request handling, structured JSON response parsing, and graceful error recovery across all API calls
- Built a stateful, component-driven UI across 5+ screens following Material Design principles — managing complex state transitions and user flows, directly transferable to React component + hooks patterns
- Designed Firebase Realtime Database schema with optimized read/write operations, reducing data fetch time by ~40% through structured indexing and query optimization — applied DBMS normalization principles
- Applied HashMap-based deduplication logic on summarized content, reducing redundant output by O(n) vs a naive $O(n^2)$ approach — demonstrating practical DSA application to a real product problem
- Owned the complete development lifecycle: requirements breakdown, pseudocode design, API integration, QA testing, and deployment — delivered independently within a 6-week timeline

Smart Home Monitoring System — IoT Backend & Real-Time Data Pipeline

Event-Driven Architecture • Firebase • Sensor API Integration • Rule-Based Automation | 2024

- Built a real-time data pipeline ingesting live streams from 3+ IoT sensors; applied event-driven architecture with threshold-based rule computation to trigger automated device responses in under 2 seconds
- Designed a rule-based computation engine evaluating multiple simultaneous sensor conditions — used priority queue logic to sequence rule evaluation for correctness and execution efficiency
- Structured Firebase database with normalized schemas and indexed queries, enabling paginated historical log retrieval across 1,000+ sensor events with consistent sub-100ms query performance
- Reduced manual device-management overhead by 60% through automated scheduling logic built using modular, testable functions — fully documented with API contracts and deployment guide
- Debugged real-time race conditions in sensor event processing using methodical root-cause analysis; documented findings and implemented a queue-based fix to guarantee event ordering

EDUCATION

B.E., Computer Science and Engineering 2023 – 2027

Sethu Institute of Technology, Kariapatti | Current GPA: 6.9 %

Higher Secondary in Computer Science 2022 – 2023

Chandler Matric Higher Secondary School, Madurai . Percentage : 57.67

CERTIFICATIONS & ACTIVE LEARNING

Cloud Computing — NPTEL (IIT Kharagpur) June 2025

Currently practicing: DSA problems (Arrays, Hashing, Linked Lists, Trees, Binary Search) on LeetCode/HackerRank; JavaScript ES6+ and React Hooks; Node.js RESTful API development; SQL query optimization

CORE STRENGTHS

DSA-First Problem Solving • Clean, Documented Code • Full Ownership Mindset • API Design & Integration • Rapid Learning • Agile Collaboration • Edge-Case Awareness